

Loan Default Prediction System

ML Internship Final Project Summary · Ahmed Tareq Rajab Bin Hadjeh

Project Goal

Build a machine learning model to predict whether a loan applicant will be approved or rejected, based on historical applicant data (income, credit history, marital status, property area, and related factors) — helping identify risk before a loan is issued.

Chosen Model

Logistic Regression was selected as the primary model. Naive Bayes performed comparably on aggregate metrics and was retained as an interpretable baseline, but Logistic Regression offers the same accuracy with directly interpretable feature coefficients, and both clearly outperformed Decision Tree and Random Forest on this dataset.

Final Accuracy

Model	Accuracy	Macro F1	Reject Precision/Recall	Approve Precision/Recall
Naive Bayes	78.9%	0.70	0.83 / 0.39	0.78 / 0.96
Logistic Regression	79.7%	0.70	0.93 / 0.37	0.78 / 0.99
Decision Tree	66.7%	0.62	0.46 / 0.50	0.77 / 0.74
Random Forest	75.6%	0.65	0.72 / 0.34	0.76 / 0.94

Logistic Regression reached 79.7% test accuracy (macro F1 0.70) — the strongest overall result, though it under-recalls the minority “Reject” class (37%), a limitation shared by all four models tested.

Insights from EDA

- Credit History is the single strongest predictor of loan approval — applicants with a positive credit history are approved at a dramatically higher rate, far more decisively than any income-related feature.
- Loan_Status is imbalanced (~2:1 approved to rejected), which explains why every model tested is better at recognizing approvals than rejections.
- Applicant/Co-applicant income and Loan Amount show only weak-to-moderate relationships with approval outcome — numerical features alone do not separate the two classes well.
- Semiurban properties have the highest approval rate among the three property area types, making it a secondary useful predictor.